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i	X_i	y
0	0	2.2500
1	0.5	
1	1	1.6420

$$y(0.5) = 0.2500 \times \frac{(0.5 - 0)}{(1 - 0)}$$

$$= 0.2500 + 0.5/1$$

$$= 0.2500 + 1.1950$$

$$y(0.5) = 1.9450$$

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Ejercicio 22 Para 22
el pago aprox: 1,540.43

Ejercicio 3 $\Rightarrow y_0 + 3\Delta y_0 + 3\Delta^2 y_0 + \Delta^3 y_0$

Ejercicio 4 $\Rightarrow y_3 = y_4 - (y_4 - y_3)$

$y_3 = y_3$

$$(x_0, y_0) = (7, 3) \quad x = 1.5$$

$$(x_1, y_1) = (2, 8)$$

$$(x_2, y_2) = (3, 15)$$

$$L_0(x) = (1.5 - 2) \times (1.5 - 3) / (1 - 2) \times (1 - 3)$$

$$L_1(x) = (1.5 - 1) \times (1.5 - 3) / (2 - 1)$$

Interpolación Cuadrática

Se dispone de 3 datos, se lleva a cabo un polinomio de 2do Orden

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = \frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$x_0 = 1 \quad 0$$

$$x_1 = 4 \quad 1.386$$

$$x_2 = 6 \quad 1.7617$$

$$\frac{1.386294 - 0}{4 - 1} = 0.46209812$$

$$\frac{1.761754 - 1.386294}{6 - 4} - \frac{1.386294 - 0}{4 - 1}$$

$$6 - 1$$

$$= -0.0518781$$